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By

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Connected and Autonomous Electric Vehicles: Estimation of Necessary Variables for Vehicle Tracking.

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Summary

The objective of the thesis is to estimate the necessary variables for tracking of connected and autonomous vehicles. The task of estimation requires the selection of suitable dynamical models upon which estimators are derived. For the sake of estimating necessary variables needed for the semi-autonomous adaptive cruise controller system subjected to cyber-attack, an unknown input observer is derived in both discrete and continuous time domains. Such observers are used to estimate unmeasured state variables when the considered system is subjected to unknown input. The existence of an unknown input observer (UIO) dictates the detectability and invertibility condition (observer matching condition), the last one is not always verified. Consequently, it is impossible to design an UIO. This problem is circumvented by including time derivative of the output to the considered system in continuous time. For this purpose, an adaptive high order sliding mode differentiator is proposed to estimate high order derivatives of the output signals. Whereas, in discrete-time the concept of differentiation does not exist. But instead, the output (as a sequence of value rather than a continuous function) is shifted in a way the OMC is verified. Finally numerical simulations are given to illustrate the effectiveness of the proposed algorithms and methods.

Keywords: Adaptive cruise controller, autonomous vehicles, vehicle dynamics, Unknown input observer, sliding modes, Identification,cyber-attack, state estimation.

Résumé

L'objectif de la thèse est d'estimer les variables nécessaires au suivi des véhicules connectés et autonomes. La tâche d'estimation nécessite la sélection de modèles dynamiques appropriés sur lesquels les estimateurs sont dérivés. Afin d'estimer les variables nécessaires pour le système de contrôleur de croisière adaptatif semi-autonome soumis à une cyber-attaque, un observateur d'entrée inconnu est dérivé dans les domaines temporels discrets et continus. De tels observateurs sont utilisés pour estimer des variables d'état non mesurées lorsque le système considéré est soumis à une entrée inconnue. L'existence d'un observateur d'entrée inconnu (UIO) dicte la condition de détectabilité et d'inversibilité (condition d'appariement de l'observateur), cette dernière n'étant pas toujours vérifiée. Par conséquent, il est impossible de concevoir un UIO. Ce problème est contourné en incluant la dérivée temporelle de la sortie vers le système considéré en temps continu. A cet effet, un différentiateur adaptatif par modes glissants d'ordre élevé est proposé pour estimer les dérivées d'ordre élevé des signaux de sortie. Alors qu'en temps discret le concept de différentiation n'existe pas. Mais au lieu de cela, la sortie (en tant que séquence de valeurs plutôt qu'en tant que fonction continue) est décalée d'une manière dont l'OMC est vérifiée. Enfin, des simulations numériques sont données pour illustrer l'efficacité des algorithmes et méthodes proposés.

Mots clés : Régulateur de vitesse adaptatif, véhicules autonomes, dynamique des véhicules, observateur à entrée inconnue, modes glissants, cyber-attaque, estimation d'état

1. Chapter1: State of art on adaptive cruise control and problem statement

1.1 Introduction

The accelerated growth of the number of cars currently in circulation around the world poses many contemporary challenges. In fact, if we go back to the seventies of the last century, particularly in 1976, we will find that the number of cars at that time did not exceed 342 million. Then, this figure increased in 1996 to reach 670 million. Subsequently, in the following years, the automotive industry experienced significant development, and the dependence on cars as the primary means of transportation became inevitable in many countries, leading to the global number of cars reaching 1.4 billion in 2020, with 39.3 million in France alone. This significant number is due to the sharp increase in the annual car production in various factories before the appearance of the coronavirus pandemic, where the average number was 90 million cars per year before this pandemic. It is expected that this number will continue to rise in the coming years, reaching 1.8 billion cars manufactured annually in 2035 and 9 billion in 2100.

This great number of cars imposes many constraints, in particular concerning traffic jam and road safety, which needs to find out compliant solutions for the intelligent future of the vehicle. One of the proposed solutions to the latter is the intelligent or autonomous convoy, the principle of which is to drive autonomously a set of cars gathered around a convoy and under supervision of a lead car which gives orders to the rest of the convoy to accomplish the task entrusted to him in the best conditions. This proposal contributes to reducing the problems related to road safety and the safety of passengers, because all the cars are all driving in parallel and at the same speed, while maintaining a previously agreed safety distance. The technology that ensures a safe driving of a vehicle in a platoon is known as Adaptive Cruise Control or ACC. While regular cruise control can maintain a steady speed that the driver sets, Adaptive cruise control is an enhancement of the conventional cruise control since it adjusts the speed of car to match the speed of the heading car. From a material perspective, ACC is a vehicle's controller that uses a number of measurements from sensors of the car like radar, camera and wheel speed sensors, communication between cars and forces applied to the vehicle. Nevertheless, sensors are not spared from neither malfunctioning nor external disturbances that ruin to their reliability and finally performance of the Adaptive Cruise Control system. One of the interesting solutions to cope with those situations is using soft sensors or "Observers" that allow to estimate signals needed for ACC system to work and to detect any external disturbances that taint the inter-vehicle communication.

1.2 Adaptive cruise control

Nowadays, cars are being increasingly equipped with adaptive cruise control (ACC). ACC enables the host vehicle to follow the longitudinal motion of the vehicle in front, which will be referred to as the target vehicle, at a desired inter-vehicle distance. This distance, in most ACC applications, depends on what is called time gap [1]. A radar system attached to the front of the vehicle detects whether slower moving vehicles are in the ACC vehicle's path. Hence, if a slower moving vehicle is detected, the ACC system will slow down the host car in order to keep the safety distance. In the same manner, If the forward car is moving faster or is outside of the ACC vehicle's path, ACC system will speed up the host car until the desired distance(safe) is assured and that the vehicle's speed is set to what the driver has ordered "Cruise Control". The control of distance is realized by a feedback controller to minimize the error between the desired and measured inter-vehicle distance as shown in the following figure:

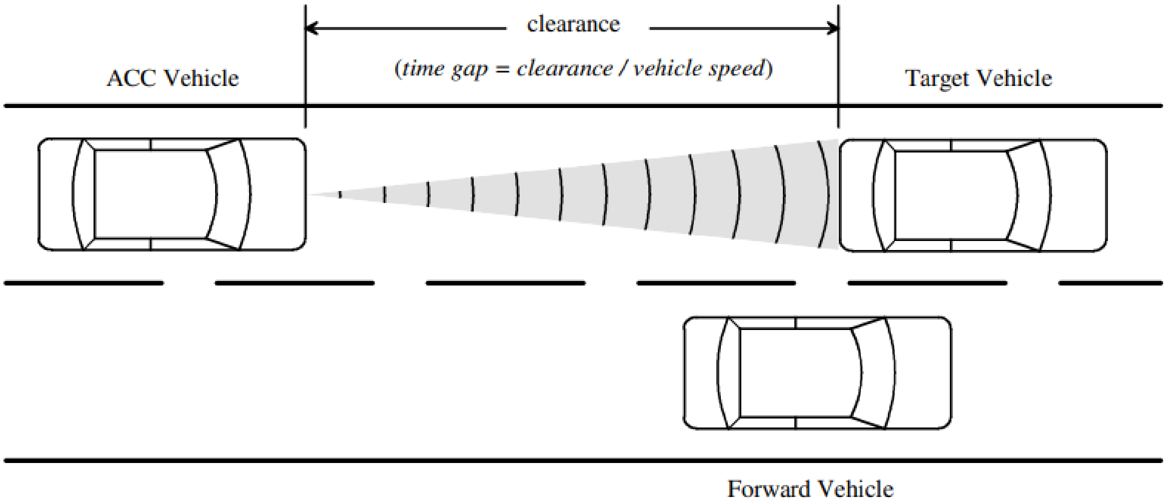

Figure 1. ACC vehicle relationships.

Another extension of ACC system is Cooperative Adaptive Cruise Control (CACC), which upgrades the functionality of ACC to automate a longitudinal motion via wireless inter-vehicle communication in a string of vehicles or platoon [2] like shown in the following figure:

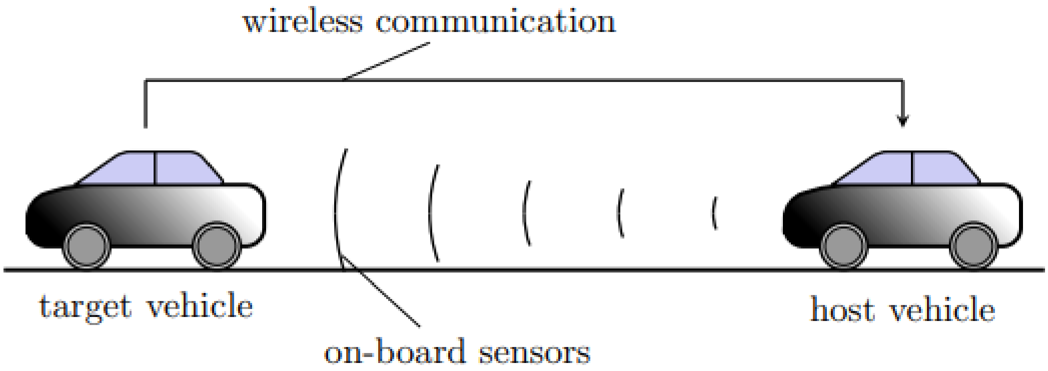

Figure 2. Illustration of CACC working on a vehicle platoon.

1.3 Challenges in ACC

It is worth to point out that autonomous cars, technically, are not all autonomous and fully self-driving and the degree of autonomous driving depends on the level and extent of automatic functionality. In fact, there are five levels of autonomous cars; cars with no autonomous features up to cars that operate entirely on their own without any driver presence. Thus, high levels of automation (namely 4 and 5) exclude the driver under precise environments (Level4) or in all possible solutions (Level5), by ensuring autonomous monitoring of the surrounding area, the performance of the sensors and the algorithms and by deciding on the actions so that the vehicle operate safely. Consequently, the monitoring of the performance of vehicle sensors is of serious concern. In other words, the higher the automation is, the more the vehicle is dependent on reliable and accurate sensors.

These challenges enclose (but not entirely) the unavailability/mediocrity of measurements and cyber-attacks [3] in the case of inter-vehicle communication [4-5]. With that being said, the detection and mitigation of cyber-attacks, sensors faults and the estimation of necessary variables are essential for operating the vehicle in a safe state.

1.4 Research objective and contribution

In the laboratory of CRAN Longwy, and especially the team *Commande et ObServation des Systèmes Non-Linéaires (COSsyNL)* has carried tasks around state estimation and cyber-attack detection, a paper published in 2019 of *Zemouche, A* [5] from the laboratory, developed a resilient controller under cyber-attacks in connected ACC vehicles. This controller incorporated cyber-attack estimate to mitigate the effect of faulty signals transmission between vehicles. An unknown input observer was developed to estimate states as well as sensor faults by incorporating adaptive law to estimate this later. This paper assumed the system to verify an essential condition that allows the application of unknown input observer, this condition is known as "Observer Matching Condition" OMC. Whereas, this is not always the case, such restriction push to find other patterns to design this observer. This restriction is overcome in the literature by introducing additional measurement so that the new system verifies the OMC.Moreover, in this paper, only the case of constant and slowly time-varying cyber-attacks was considered. Whereas, this assumption is not always true covers since attacks and faulty signals can be time-varying, fast or stochastic.

- **Contribution:**

The contribution of this thesis is to apply another type of observers and adapt them to the problematic under consideration. Briefly, in this thesis we have:

1. Developed a discrete-time Unknown Input Observer for Semi-Autonomous adaptive cruise control problem.

2. Developed an Adaptive High Order Sliding Mode Differentiator in order to estimate high order derivatives of the output for Semi-Autonomous adaptive cruise control problem.

1.5 Outline

The remainder of this thesis is organized as follows. Chapter 2 is dedicated to modeling of Semi-Autonomous adaptive cruise control system. In chapter 3, the observer design is carried out and the implementation of the observer as well. Chapter 4 presents simulation results of applied estimation techniques.

2. Chapter2: Modelling of Semi-Autonomous Adaptive Cruise Control system

2.1 Autonomous vehicles:

A self-driving car is capable of sensing the surrounding environment and navigation without the human intervention to accomplish a task. The car is equipped with a number of instruments such as: GPS unit, inertial navigation system and a range of sensor including laser, finder radar and video.

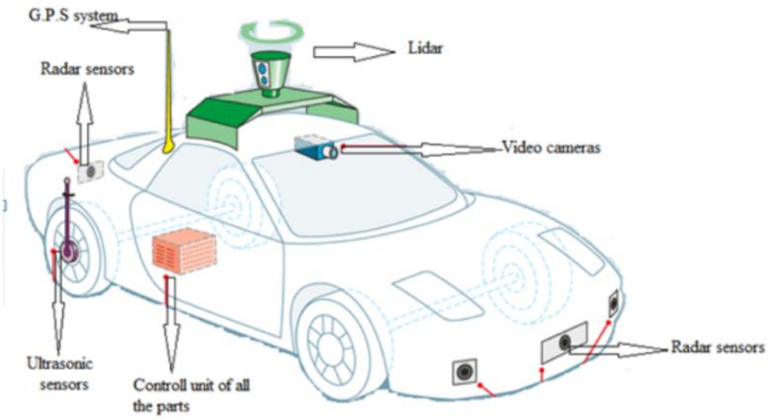

Figure 3. Instruments of a typical autonomous vehicle.

Employing the map, the vehicle seeks the most efficient route towards its destination while circumventing obstacles among several available paths. Once the optimal paths are identified, the vehicle processes this information into actionable commands, which are then transmitted to its actuators. These actuators, responsible for steering, braking, throttle, and other functions, execute the commands accordingly.

On a standard autonomous vehicle, we find the following sensors:

1- Lidar: Similar to radar but uses lasers to measure distance instead of sound, with a range of up to 200m.
2- Video Cameras: Records multiple still shots (frames) to capture a 2-dimensional motion picture.
3- Ultrasonic Sensors: Identify objects close to the vehicle and support autonomous drive-in low speeds.
4- Radar: Uses bursts of sound to measure distance by measuring the time it takes for the sound to return to the sensor.
5- GPS system: Uses real time geographical data received from several GPS satellites to calculate longitude, latitude, speed and course to help navigate a car.

2.2 Semi-Autonomous Adaptive Ccruise Control model(SA-ACC)

2.2.1 Definition

A semi-autonomous adaptive cruise control system offers substantial benefits compared to existing adaptive cruise control systems. This innovative system enhances highway safety and traffic flow by combining the advantages of autonomous vehicles and fully automated highway systems, allowing vehicles to cooperate as a platoon. Unlike traditional platoon systems, these semi-autonomous systems can be readily implemented on current highways, where both manually driven and adaptive cruise controlled vehicles can coexist harmoniously.The eminent feature of SA-ACC system is allowing inter-vehicle communication which improves string stability[21].

The basic control algorithm idea of autonomous vehicles is based on the choice of a speed set point the driver wants, an upper controller receives the actual speed of the vehicle and produces the desired acceleration for the vehicle. Due to the hierarchical strucutre of ACC control systems, lower controller receives the disered acceleration and dertermines the throttle input required to track the desired acceleration.

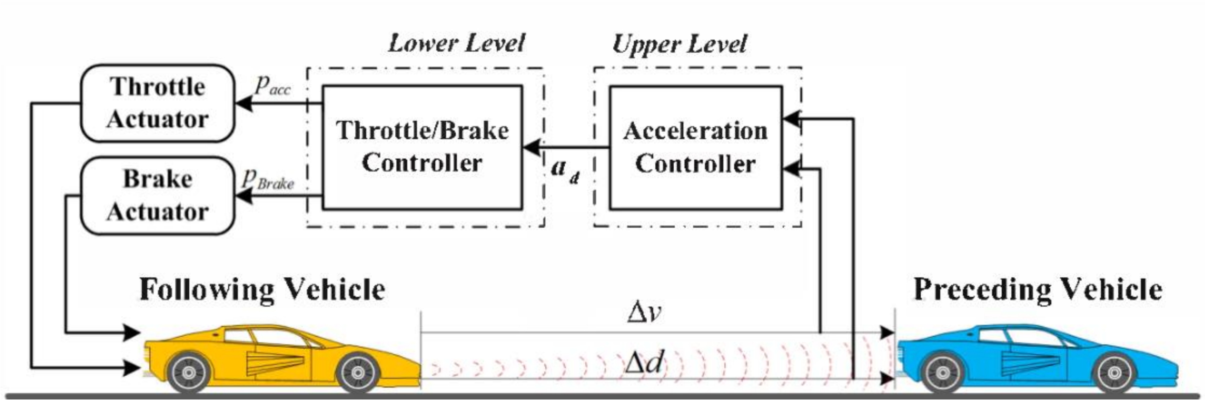

Figure 4. The hierarchical control framework of ACC system.

2.2.2 SA-ACC system

The SA-ACC system aims to obtain high traffic capacity and small inter-vehicle spacing while using communication from only the preceding vehicle on the highway. The controller of the SA-ACC system is based on the time gap or clearance spoken of in the introduction. The time gap is usually referred to as spacing policy, we present two types of spacing policies:

- Constant Spacing Policy :

$$d_{des} = L \tag{2.7}$$

d_{des} denotes the desired inter-vehicle spacing, and L represents a fixed positive constant

- Constant time gap spacing policy:

$$d_{des} = L + h\dot{x} \tag{2.8}$$

L in a constant, h is time gap and $\dot{x}$ is the velocity of the vehicle hosting the ACC system.

The design of SA-ACC system is based on the second spacing policy. In the presence of actuator dynamics represented by a first-order lag, the vehicle model of the SA-ACC vehicle is described as:

$$\tau\dddot{x}_i + \ddot{x}_i = u_{syn} \tag{2.9}$$

We define the spacing policy of SA-ACC as :

$$\bar{\varepsilon_i} = \varepsilon_i + h\dot{x}_i \tag{2.10}$$

$$\varepsilon_i = x_i - x_{i-1} + L_i \tag{2.11}$$

Where τ is a lag constant associated with the lower dynamics of the vehicle(lower level), x_i is the i^{th} vehicle position, and u_{syn} is the control input of the i^{th} vehicle. According to [5],[19], the control law u_{syn} is given as follows:

$$u_{syn} = -k_1\ddot{x}_{i-1} + (k_1 + hk_1k_2)\ddot{x}_i - \frac{1}{h}(1 - hk_1k_2)\dot{\varepsilon}_i - \frac{k_2}{h}\varepsilon_i - k_2\dot{x}_i \tag{2.12}$$

Where $\ddot{x}_{i-1}$ is the acceleration of the preceding vehicle obtained by using the inter-vehicle communication. k_1, k_2 are controller design parameters. Detailed procedure to obtain the controller and determine its parameters can be found in [21].

2.2.3 New model for SA-ACC system

In this section, we consider the example of two vehicle noted as leading for the red car and following for the blue car, the one that is hosting the SA-ACC system.

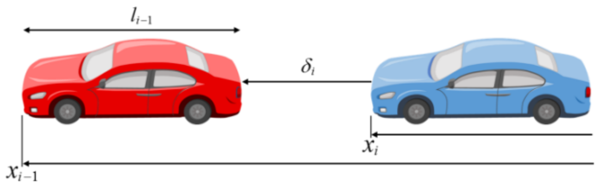

Figure 5. Preceding and following vehicles.

The model of this new system is built upon the spacing error between the two vehicles. The radar measuring the distance between vehicles is defined as:

$$\delta_i = x_{i-1} - x_i - l_{i-1} \tag{2.13}$$

Time differentiations of δ_i yields :

$$\dot{\delta}_i = \dot{x}_{i-1} - \dot{x}_i \tag{2.14}$$

$$\ddot{\delta}_i = \ddot{x}_{i-1} - \ddot{x}_i \tag{2.15}$$

$$\dddot{\delta}_i = \dddot{x}_{i-1} - \dddot{x}_i \tag{2.16}$$

By using the equations 2.13 to 2.16, the controller becomes:

$$u_{syn} = (hk_1k_2)\ddot{x}_{i-1} - k_2\dot{x}_i - (k_1 + hk_1k_2)\ddot{\delta}_i + \frac{1}{h}(1 - hk_1k_2)\dot{\delta}_i + \frac{k_2}{h}\delta_i - \frac{k_2}{h}(L_i - l_{i-1}) \tag{2.17}$$

Concerning the preceding vehicle,the associated time lag constant does not need to be known, since we obtain the acceleration information directly using wireless communication. Thus, without considering its lower order dynamics, the preceeding vehicle model can be represented as:

$$\begin{gathered}\ddot{x}_{i-1} = a_{i-1} \\ \dddot{x}_{i-1} = 0\end{gathered} \tag{2.18}$$

Before writing the equtions describing the spacing error dynamics, we assume that a the transmission of a_{i-1} is not safe but is corrupted by a cyber-attack, this latter can be a false information entering the same communication channel as the real acceleration a_{i-1}, or a transmission delay of the information, we can write :

$$\mu = a_{i-1} + f_c \tag{2.19}$$

Combining the equations 2.13 to 2.16 and 2.18,2.19, we write the spacing error dynamics :

$$\begin{bmatrix} \dot{\delta}_i \\ \ddot{\delta}_i \\ \dddot{\delta}_i \end{bmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \dfrac{-k_2}{\tau h} & \dfrac{-k_3}{\tau h} & \dfrac{k_1 - k_3}{\tau} \end{pmatrix} \begin{bmatrix} \delta_i \\ \dot{\delta}_i \\ \ddot{\delta}_i \end{bmatrix} + \begin{pmatrix} 0 \\ 0 \\ \dfrac{k_2}{\tau h} \end{pmatrix} \dot{x}_i + \begin{pmatrix} 0 \\ 0 \\ \dfrac{k_3}{\tau h} \end{pmatrix} \mu + \begin{pmatrix} 0 \\ 0 \\ \dfrac{k_2 (L_i - l_{i-1})}{\tau h} \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ \dfrac{k_3 - k_1}{\tau} \end{pmatrix} f_c \tag{2.20}$$

With $k_3 = 1 - h k_1 k_2$.

3. Chpater3: Observer Design of Semi-Autonomous Adaptive Cruise Control system

3.1 State Observer for dynamical systems

3.1.1 Principle of state estimation

An observer consists of an auxiliary dynamical system, whose inputs are the measured inputs/outputs of a system, and the outputs are assumed to provide an estimation of its state, as depicted below.

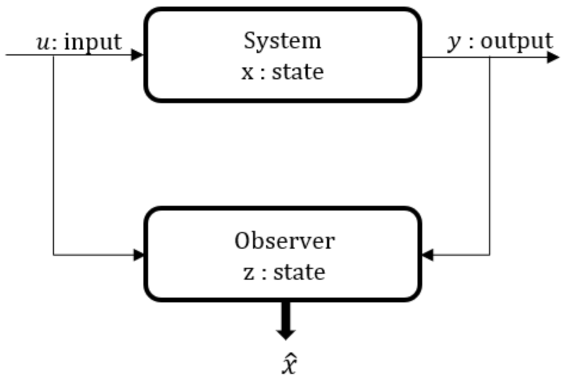

Figure 6. Principle of state estimation.

3.1.2 Families of state observers for dynamical systems

In general, the state observers are grouped in three families:

- Observers for linear systems:

State observers for linear systems are estimators designed for linear systems or, at the limit, linearized systems around an operating point. Luenberger observer is the simplest type of estimators introduced by D. Luenberger[7]. Although these observers are straightforward to design and implement, they are highly vulnerable and fail to accurately observe the states when the system exhibits non-linearities, significant deviations from the operating point, parametric variations, measurement noise, parametric uncertainties, or modelling errors.

- Observers for nonlinear systems:

Unlike linear systems, until today, there is no systematic method for constructing observers for non-linear systems. Nonlinear systems are diverse and fall into different classes. For some classes, non-linear observers can be designed, while for others, they may not be feasible. Many existing techniques are used, we list some of them: globally linearizing observers [9], extended Luenberger observer [10], high gain observer [11], discontinuous observers [12].

- Disturbance observers

Physical systems are often subject to unknown inputs, which makes the use of classical observers impossible. This is where disturbance observers stand out from the other observers mentioned earlier. They allow for the estimation of both the system state and the unknown disturbances. However, these types of observers operate under certain mathematical conditions and structural properties that are difficult to verify [13]

3.2 State-space representation of SA-ACC model and initiation to Unknown Input Observer

This section is devoted to represent to SA-ACC system in the state-space form. Next the concept of unknown input observer, existence condition and verification are presented.

3.2.1 State space representation of the SA-ACC system

Let us consider the equation (2.20) in the following form:

$$\begin{cases} \dot{x} = Ax + Bu + F\mu - Wf_c + \Delta \\ y = Cx \end{cases}, \tag{3.1}$$

where

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{-k_2}{\tau h} & \frac{-k_3}{\tau h} & \frac{k_1 - k_3}{\tau} \end{pmatrix}, B = \begin{pmatrix} 0 \\ 0 \\ \frac{k_2}{\tau h} \end{pmatrix}, F = \begin{pmatrix} 0 \\ 0 \\ \frac{k_3}{\tau h} \end{pmatrix}, W = \begin{pmatrix} 0 \\ 0 \\ \frac{k_3 - k_1}{\tau} \end{pmatrix}$$

$$\Delta = \begin{pmatrix} 0 \\ 0 \\ \frac{k_2(L_i - l_{i-1})}{\tau h} \end{pmatrix}, C = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

$x(t) \in \mathbb{R}^3$ is the state variable, $u(t), v(t)$ and $f_c \in \mathbb{R}$ is the system vector inputs and fault respectively,$y(t) \in \mathbb{R}^2$ is the system output.

3.2.2 Unknown input observer design of linear-time invariant systems

As we have discussed previously, the SA-ACC can be vulnerable to cyber-attack that are most of the time unknown. From dynamic system perspective, this is considered as an unknown input. The synthesis of observers needs the accessibility to all inputs subjecting the system being observed, if at least one input is not measured, the observer cannot be constructed.

Fortunately, there exists other type of observers that can estimate, under some conditions that will be shown later, states even when the input is not known. Such observers are called Unknown Input Observers (UIO).

Let us consider a simple linear time-invariant system:

$$\begin{cases} \dot{w}(t) = Gw(t) + Er(t) \\ y(t) = Cw(t) \end{cases}, \tag{3.2}$$

with w is state, r is unknown input and y is the output.

An observer is defined as an unknown input observer, for the system described by (3.2), if its state estimation error vector, $e(t)$, defined as:

$$e(t) = x(t) - \hat{x}(t) \tag{3.3}$$

approaches zero asymptotically regardless of the presence of the unknown input $r(t)$, in the system. The structure of the UIO observer is given by

$$\begin{cases} \dot{z}(t) = Nz(t) + Mr(t) + Ly(t) \\ \hat{x}(t) = z(t) + Ky(t) \end{cases}, \tag{3.4}$$

where N, M, L and K are observer matrices designed so that the estimation error (3.3) tends to zero. The details of the matrices computation is provided in [14].

3.2.3 Conditions for existence of an unknown input observer

The necessary and sufficient conditions for the system in (3.2) to have an UIO (3.4) are:

1) $rank(CE) = rank(E)$
2) (C, G) is detectable

See appendix for more details about detectability condition.

3.3 Unknown input observer design for SA-ACC system

In the previous section, the SA-ACC system is subjected to unknown input or cyber-attack. So far, our objective is to design the UIO in order to estimate states and reconstruct the unknown input to be compensated.

To do so, we must verify the condition 1 and 2.

3.3.1 Verification of the existence conditions

- First condition:

For detectability, we consider the matrix $\begin{bmatrix} sI_3 - A & W \\ C & 0 \end{bmatrix}$, we compute the rank of this matrix for all $s \in C, Re(s) \geq 0$. Numerical values of matrices are given in the appendix.

$rank \begin{bmatrix} sI_3 - A & W \\ C & 0 \end{bmatrix} = rank \begin{bmatrix} s & -1 & 0 & 0 \\ 0 & s & -1 & 0 \\ 12.5 & 10 & s+7 & 7 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} = (-1)(-1)(-7) \neq 0$, we conclude that $\begin{bmatrix} sI_3 - A & W \\ C & 0 \end{bmatrix}$ is of full rank $\forall s \in \mathbb{C}, Re(s) \geq 0$. Therefore, the system (3.1) is detectable.

- Second condition:

The second condition concerns the satisfaction of $rank(CW) = rank(W)$, this condition is very important in the sense that it allows to inverse the matrix CW to reconstruct the unknown input. It is commonly known as **Observer Matching Condition (OMC).**

We have, $CW = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ \frac{k_3 - k_1}{\tau} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, we conclude that the observer cannot exist for the system (3.1).

To overcome the rank condition that is not satisfied, it is necessary to have additional measurement. In fact, adding a new measurement will change the output matrix C so that the rank condition is satisfied. We can notice that a third row is needed to avoid $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ since only the third component of the matrix W is not null.

3.3.2 Incorporating auxiliary output to original system

As explained above, an additional(auxiliary) information is needed to overcome the rank restriction. We simply need the relative acceleration $\ddot{\delta}_i$ information. That is why, in continuous-time, it is necessary to add another observer to estimate the derivative of the velocity $\dot{\delta}_i$. Typically, this observer is named a differentiator since it permits to provide successive derivatives of the output.

Whereas in discrete-time it consists of shifting backwards the output measurement. This leads to considering delayed output as output at the instant and the actual (real-time output) as future output. This intuitively is equivalent to the continuous case where a 'future' or output derivative is needed. Therefore, in discrete-time, we will not need a differentiator.

The fig shows the two techniques we are going to pursue. Case (1) is considered first in discrete-time, next we will treat the continuous-time case (2).

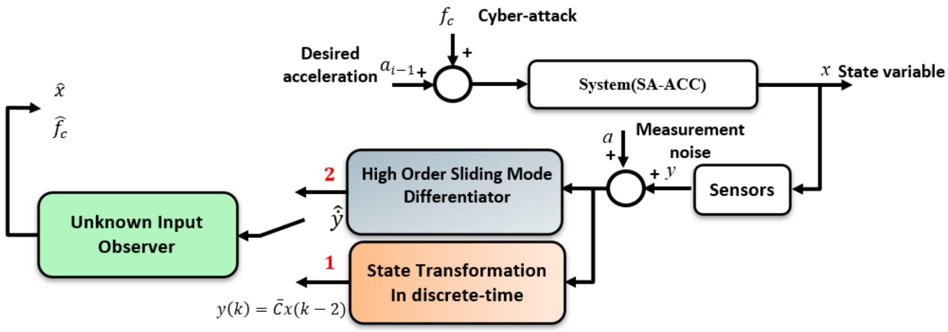

Figure 7. Principle of the proposed method of states and unknown input estimation.

3.3.3 Unknown input observer design with delayed measurement

The aim of this section is to design an unknown input observer when the OMC is not satisfied. Using Euler forward approximation, the equivalent discrete-time model of the system (3.1) is given by:

$$\begin{cases} x_{k+1} = A_d x_k + B_d u_k + F_d \mu_k - W_d f_{c\,k} + \Delta_d \\ y_k = C_d x_k \end{cases}, \tag{3.5}$$

with $A_d = (I_3 + T_s A)$, $B_d = T_s B$, $F_d = T_s F$, $W_d = T_s W$, $\Delta_d = T_s \Delta$, $C_d = T_s C$,and $T_s$being the sampling time.

From 3.5, we write:

$$y_k = C_d(A_d x_{k-1} + B_d u_{k-1} + F_d \mu_{k-1} - W_d f_{c\,k-1} + \Delta_d), \tag{3.6}$$

we can easily verify that: $C_d B_d = C_d F_d = C_d W_d = C_d \Delta_d = 0$, hence $y_k = C_d A_d x_{k-1}$.

If we consider the change of variables: $z_k = x_{k-1}$, $\bar{u}_k = u_{k-1}$,$\bar{\mu}_k = \mu_{k-1}$, $v_k = f_{c\,k-1}$ and $\bar{C}_d = C_d A_d$, the system (3.5) is expressed as:

$$\begin{cases} z_{k+1} = A_d z_k + B_d \bar{u}_k + F_d \bar{\mu}_k - W_d v_k + \Delta_d \\ y_k = \bar{C}_d z_k \end{cases}, \tag{3.6}$$

now we will verify the UIO existence condition namely OMC as follows:

$$\bar{C}_d W_d = C_d A_d W_d = \begin{pmatrix} T_s & 0 & 0 \\ 0 & T_s & 0 \end{pmatrix} \begin{pmatrix} 1 & T_s & 0 \\ 0 & 1 & T_s \\ \frac{-T_s k_2}{\tau h} & \frac{-T_s k_3}{\tau h} & \frac{T_s(k_1 - k_3)}{\tau} \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ \frac{T_s(k_3 - k_1)}{\tau} \end{pmatrix}$$

$$\bar{C}_d W_d = \begin{pmatrix} 0 \\ 0 \\ \frac{T_s^3(k_3 - k_1)}{\tau} \end{pmatrix},$$

therefore, $rank(\bar{C}_d W_d) = rank(W_d)$ when $T_s \neq 0$. Finally, we conclude that there exists an unknown input observer for the system (3.6) if it is detectable(condition1).

Let us consider the change of variable: $\xi_k = [z_k^T \ \ v_k]^T$, We obtain the following discrete-time linear descriptor system:

$$\begin{cases} E\xi_{k+1} = A_\xi \xi_k + B_\xi \bar{u}_k + F_\xi \bar{\mu}_k + \Delta_\xi \\ y_k = C_\xi \xi_k \end{cases}, \tag{3.7}$$

with $E = [I_3 \ W_d]$, $A_\xi = [A_d \ 0]$, $B_\xi = [B_d^T \ 0]^T$, $F_\xi = [F_d^T \ 0]^T$, $\Delta_\xi = [\Delta_d^T \ 0]^T$and $C_\xi = [\bar{C}_d \ 0]$.

Now let us consider the following observer structure to estimate states and unknown input simultaneously:

$$\begin{cases} w_{k+1} = Nw_k + Ly_k + M\bar{u}_k + G\bar{\mu}_k + \ P_z \Delta_\xi \\ \hat{\xi}_k = w_k + Q_z y_k \end{cases} \tag{3.8}$$

The matrices N, L, M, G, P_z, Q_z are observer's design parameters chosen such that the estimation error $\tilde{\xi}_k = \xi_k - \hat{\xi}_k$ converges to zero. One can refer to appendix for more details about this. The observer gain is computed based on solving the proposed Linear Matrix Inequality (LMI), see appendix.
The simulation results are presented in the next chapter.

3.3.4 Design of Adaptive High Order Sliding Mode Differentiator

This section is dedicated to design an observer that estimates real-time derivatives of the system 3.1. those estimates are called auxiliary outputs. Such outputs are introduced so that OMC is satisfied with respect to the unknown input. This an alternative to the earlier-mentioned method.

Let us consider the system (3.1), when the OMC is always not satisfied

$$\begin{cases} \dot{x} = Ax + Bu + F\mu - Wf_c + \Delta \\ y = Cx \end{cases},$$

the following hypothesis is assumed.

- States, unknown input and their derivatives are bounded by some constants.

We consider the output $y_a = C_a x$ where $C_a = [\ c_1 \ c_1 A \ldots c_1 A^{\gamma_1 - 1}, \ldots, c_p c_p A \ldots c_p A^{\gamma_p - 1}]^T$ with p is the number of outputs and γ_i is the relative degree with respect to the unknown input i.e., number of times the outputs must be differentiated to have the unknown input explicitly appears [26].We have $p = 2$ hence $y = [y_1, y_2]^T$, $C = [c_1, c_2]^T$and next we compute the relative degrees γ_1 and γ_2.

- First output: $y_1 = c_1 x$

We differentiate with respect to time, we obtain:

$$\dot{y}_1 = c_1 \dot{x} \tag{3.9a}$$
$$\dot{y}_1 = c_1(Ax + Bu + F\mu - Wf_c + \Delta) = c_1 Ax \tag{3.9b}$$
$$\ddot{y}_1 = c_1 A^2 x \tag{3.9c}$$
$$\dddot{y}_1 = c_1 A^3 x + c_1 A^2 Bu + c_1 A^2 F\mu - c_1 A^2 W f_c + c_1 A^2 \Delta, \tag{3.9d}$$

hence $\gamma_1 = 3$

- Second output: $y_2 = c_2 x$

We differentiate with respect to time, we obtain:

$$\dot{y}_2 = c_2 \dot{x} \tag{3.10a}$$
$$\dot{y}_2 = c_2(Ax + Bu + F\mu - Wf_c + \Delta) = c_2 Ax \tag{3.10b}$$
$$\ddot{y}_2 = c_2 A^2 x + c_2 ABu + c_2 AF\mu - c_2 AW f_c + c_2 A\Delta, \tag{3.10c}$$

hence $\gamma_2 = 2$

The total relative degree $\gamma = \gamma_1 + \gamma_2 = 3 + 2 = 5$

Therefore, the augmented output $y_a = [\, c_1^T \ (c_1 A)^T \ (c_1 A^2)^T, c_2^T \ (c_2 A)^T]^T x$, the objective now is to provide the estimation of y_a through a sliding mode differentiator, we denote the estimate of y_a as

$$y_s = [\, c_1^T \ (c_1 A)^T \ (c_1 A^2)^T, c_2^T \ (c_2 A)^T]^T \hat{x} \tag{3.11}$$

As $\gamma = n + p = 5$, the state estimates can be obtained simply as $\hat{x} = C_a^{-1} y_a$ only based on the high-order sliding mode differentiator. Which means, we only need a sliding mode differentiator to estimate the states and reconstruct the unknown input.

Remark: If $\gamma < n + p$ a reduced order UIO is used [15],[16].

We consider now the auxiliary output, we can write $y_a = [\, {y_{a1}}^T, {y_{a2}}^T]^T$ where:

$$y_{a1} = [\, c_1^T \ (c_1 A)^T \ (c_1 A^2)^T]^T x = [y_{a11}, y_{a12}, y_{a13}]^T \tag{3.12}$$
$$y_{a2} = [\, y_{a21}, y_{a22}]^T x \tag{3.13}$$

Taking the equation 3.12, the state space equation is given by:

$$\begin{cases} \dot{y}_{a1} = \Lambda_1 y_{a1} + H_1 u + E_1 \mu + \Delta_1 + f_1(x, f_c) \\ y_1 = c_1 x \end{cases}, \tag{3.14}$$

where

$$\Lambda_1 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, H_1 = \begin{pmatrix} 0 \\ 0 \\ c_1 A^2 B \end{pmatrix}, E_1 = \begin{pmatrix} 0 \\ 0 \\ c_1 A^2 F \end{pmatrix}, \Delta_1 = \begin{pmatrix} 0 \\ 0 \\ c_1 A^2 \end{pmatrix} \Delta \text{ and}$$

$$f_1(x, f_c) = c_1 A^3 x - c_1 A^2 W f_c,$$

we denote $y_{a14} = f_1(x, d)$, the state equation of the system 3.14 can be then expanded to:

$$\begin{cases} \dot{y}_{a11} = y_{a12} \\ \dot{y}_{a12} = y_{a13} \\ \dot{y}_{a13} = y_{a14} + c_1A^2Bu + c_1A^2F\mu + c_1A^2\Delta \\ \dot{y}_{a14} = \dot{f}_1(x, f_c) \end{cases}, \tag{3.15}$$

we know that $\dot{f}_1(x, f_c)$ is unknown but bounded by some constant [15],[16],[17] because of the assumption above.

In order to exact estimate not only y_{a1} but also $\dot{y}_{a1}$in a finite-time the following high order sliding mode differentiator is proposed by [17] and [18].

$$\begin{cases} \dot{\hat{y}}_{a11} = \hat{y}_{a12} - w_{11} - K_{11}s_{11} \\ \dot{\hat{y}}_{a12} = \hat{y}_{a13} - w_{12} - K_{12}s_{12} \\ \dot{\hat{y}}_{a13} = \hat{y}_{a14} + c_1A^2Bu + c_1A^2F\mu + c_1A^2\Delta - w_{13} - K_{13}s_{13} \\ \dot{\hat{y}}_{a14} = -w_{14} \end{cases}, \tag{3.16}$$

with

$$\begin{cases} s_{11} = w_{10} = \hat{y}_{a11} - y_1 \; and \; s_{1i} = \hat{y}_{a1i} - \dot{\hat{y}}_{a1,i-1} \;\; i = 2, \dots, \gamma_1 \\ w_{1j} = \hat{\lambda}_{1j}\left|w_{1,j-1}\right|^{\frac{\gamma_{1-j+1}}{\gamma_{1-j+2}}} sign\left(w_{1,j-1}\right) \; and \quad j = 1, \dots, \gamma_1 + 1 \end{cases}, \tag{3.17}$$

$\hat{y}_{a1j}$ being the estimates of y_{a1j}, $(j = 1, \dots, \gamma_1 + 1)$, $K_{1i} > 0$ $(i = 2, \dots, \gamma_1)$ positive constants and $\hat{\lambda}_{1j}$ are positive scalar gains. The error dynamic system between the system's auxiliary outputs (3.15).

$$\begin{cases} \dot{e}_{a11} = e_{a12} - w_{11} \\ \dot{e}_{a12} = e_{a12} - w_{12} \\ \dot{e}_{a13} = e_{a13} - w_{13} \\ \dot{e}_{a14} = -\dot{f}_1(x, f_c) - w_{14} \end{cases}, \tag{3.18}$$

where $e_{a1j} = \hat{y}_{a1j} - y_{a1j}$. We can show that a sliding mode appears on the manifold $e_{a1j} = 0$ in finite-time if λ_{1j} are chosen properly. An adaptive approach of tuning λ_{1j} is detailed in the appendix with estimation error convergence demonstration.

Since the high order sliding mode differentiator provides also the derivative of the auxiliary output in finite time, we can take advantage of these estimated states to reconstruct the unknown input. In fact, if we consider the following variable:

$$\hat{\xi}_a = \hat{y}_{a14} + c_1A^2Bu + c_1A^2F\mu + c_1A^2\Delta, \tag{3.19}$$

Also, we know that $\hat{y}_{a14} = f_1(\hat{x}, \hat{f}_c)$= $c_1A^3\hat{x} - c_1A^2W\hat{f}_c$,and thanks to the verified OMC condition the matrix c_1A^2W is invertible, therefore the estimation of the unknown input $\hat{f}_c$ is deduced:

$$\hat{f}_c = (c_1A^2W)^{-1}\left(-\hat{\xi}_a + c_1A^3\hat{x} + c_1A^2Bu + c_1A^2F\mu + c_1A^2\Delta\right), \tag{3.20}$$

the same procedure is applied to the second output y_2. We perceive during the development that it is not necessary to apply the same procedure since y_2 is nothing but the first derivative of y_1.

With this, we do not need a sensor to measure the relative velocity since the differentiator is capable of providing it through a sensor for relative position.

4. Chapter4: Simulation results

4.1 Design parameters and calibration

In this chapter, the results obtained from simulations carried out on MATLAB are presented.

The parameters' values of the system (SA-ACC) and its controller are displayed in the table

Parameter	Value	Unit
τ	0.4	sec
L_i	7.3	m
$l_i(m)$	5	m
h	0.5	sec
T_s	0.01	sec

Table 1. Parameters of the SA-ACC system

Gains	Value
k_1	-0.8
k_2	2.5
k_3	2

Table 2. SA-ACC controller's gains

The host vehicle should pursue the following desired trajectory:

$$a_{i-1}(t) = 3sin\left(\frac{2\pi}{10}t\right), \tag{4.1}$$

we have tested the ensuing configurations of cyber-attack:

Constant attack signal

$$f_c(t) = \begin{cases} 0 & 0 < t < 2 \\ 5 & otherwise' \end{cases} \tag{4.2}$$

Time-varying attack signal

$$f_c(t) = \begin{cases} 0 & 0 < t < 4 \\ 10\sin(5\pi t)e^{-0.01.t} & 4 \leq t < 8 \\ e^{-0.02\sin(5\pi t)t} & 9 \leq t < 16' \\ 0 & 16 \leq t < 20 \end{cases} \tag{4.3}$$

4.2 Simulation results

The following section details application results to different scenarios of cyber-attack and the state estimation at each case.

4.2.1 Unknown Input observer for discrete-time linear descriptor system

State estimation: From the fig, we see that the observer is able to produce good estimation of the position, velocity and acceleration. We perceive as well the effect of unknown input compensation, the ripples in velocity and acceleration from 4 to 8sec.

Unknown input reconstruction and transmitted acceleration: In both cases (constant and time varying) fault, the observer succeeded to reconstruct the attack. But after two samples since the observer design was based upon the linear descriptor system (3.7) which is a two samples-delayed version of the system (3.5). The figure shows the difference between real and faulty transmitted acceleration, even when the signal is false, the SA-ACC control system is immune against and this thanks to the observer.

One problem with the observer is the transient phase of the cyber-attack estimation. This undesired phenomenon can be reduced either by reducing the observer gain (increase robustness) or adding a saturation.

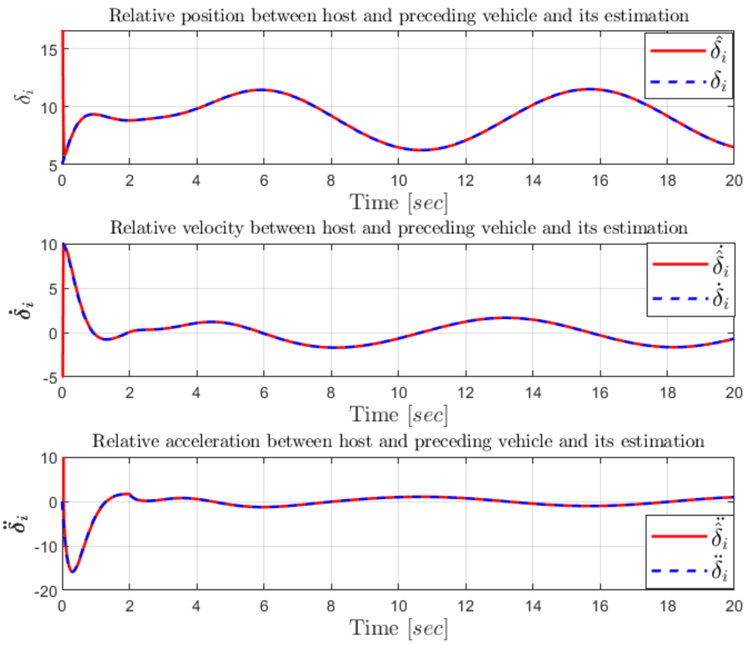

Figure 8. UIO state estimation under constant attack.

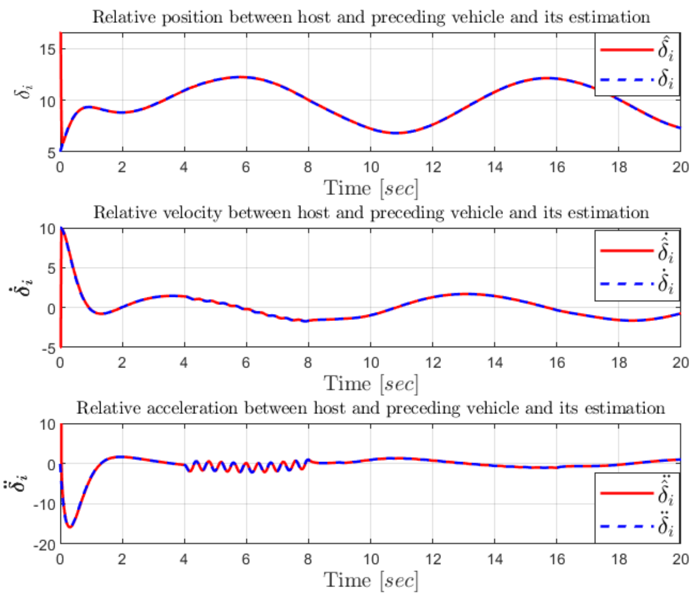

Figure 9. UIO State estimation under time-varying attack.

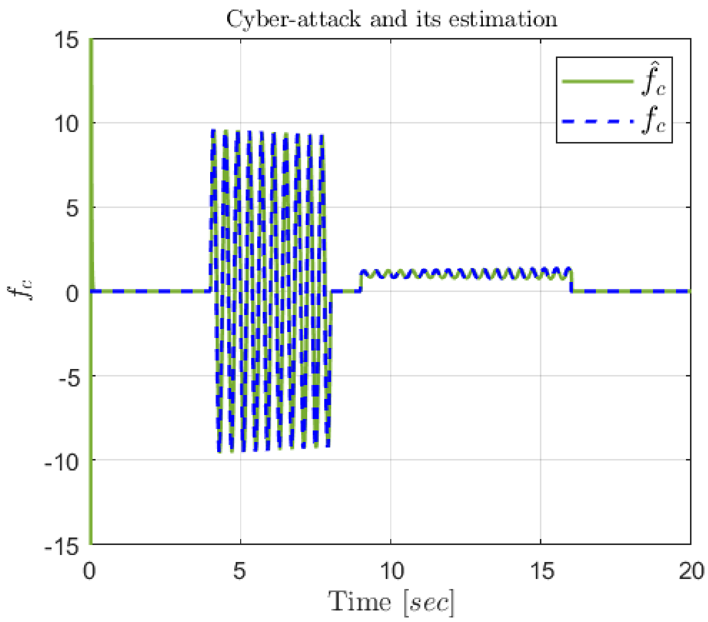

Figure 11. Time-varying attack estimation by UIO.

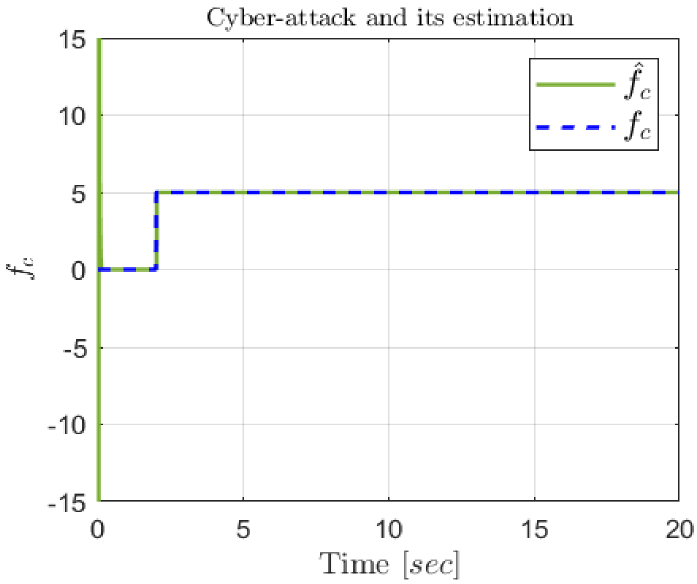

Figure 10. Constant attack estimation by UIO.

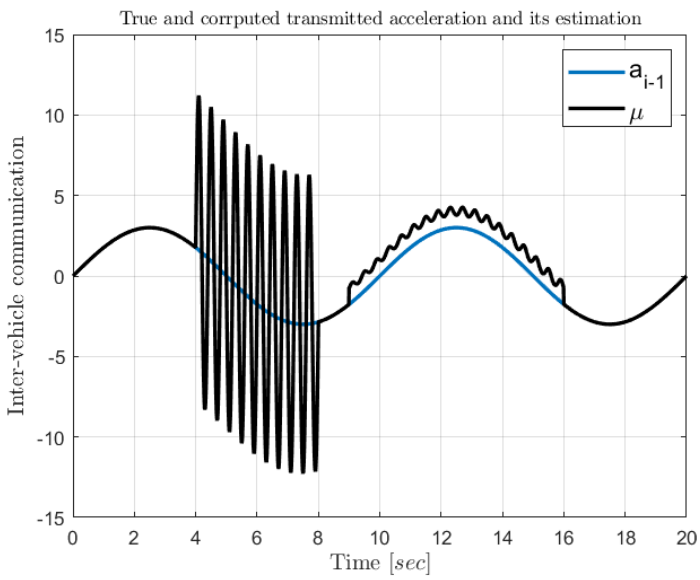

Figure 13. True acceleration information blended with time-varying attack.

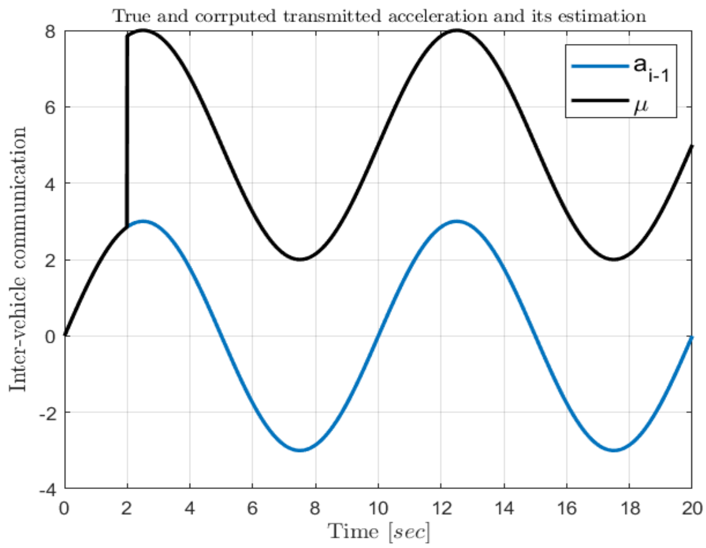

Figure 12. True acceleration information blended with constant attack

4.2.2 Adaptive high order sliding mode differentiator

In this part, we will yield the simulation results of the sliding mode differentiator that acts as an unknown input observer. Under same parameters (system and controller). The simulation results are presented below.

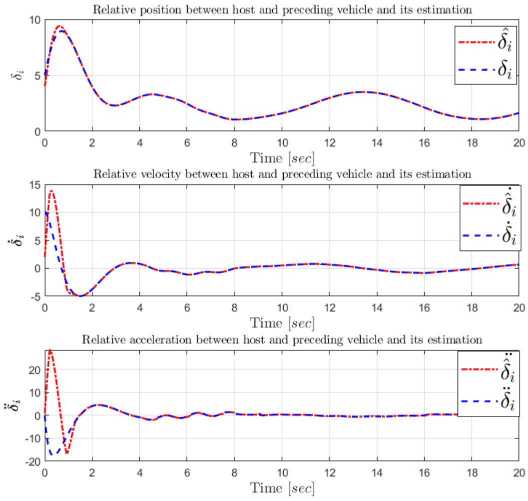

Figure 15. State estimation with adaptive high order sliding mode differentiator.

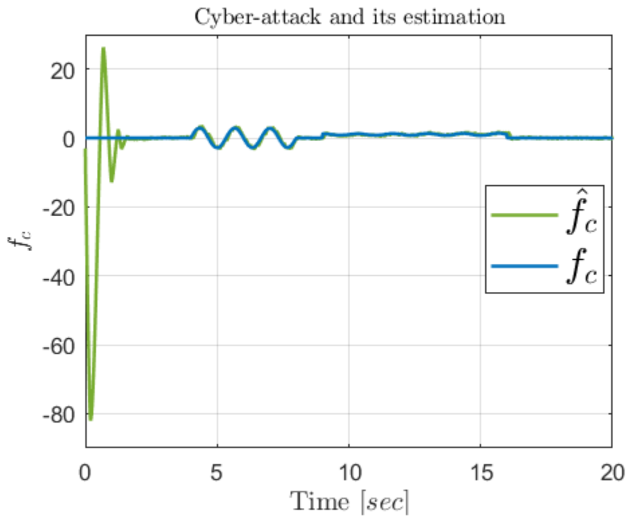

Figure 14. Cyber-attack estimation by adaptive high order sliding mode differentiator.

From the fig, we can see that the sliding mode differentiator has accurately estimated the three states. The same for fault reconstruction. Nevertheless, we observe an aggressive transient. This is a known phenomenon called "peaking". In the literature there exists an algorithm that permits to reduce even eliminate the high peak during the transient phase[16]. Otherwise, this is not an issue since the observer would start estimating the same time the vehicle is starting. If the fault is not occurred right the moment the vehicle turns on, then the observer will estimate fault without lag.

Conclusion

In this work, a relevant problem concerning the autonomous vehicle tracking has been treated. The adaptive cruise control system depends on the available measurements which need to be accurate as well as reliable. Firstly, a mathematical model of the vehicle was developed. Secondly, the problem restriction of OMC was highlighted and its consequences on observer design was pointed out. To circumvent this restriction, two methods were presented, the first one consisted in discretizing the continuous-time system and shift the output in terms of time-steps so a new output of the same system is generated. This output made the OMC condition verified. Next, the unknown input was introduced as state variable into a discrete-time linear descriptor system. After proving the detectability condition, a Lyapunov analysis of the estimation error was detailed. The resulting Lyapunov equation was expressed as a Linear Matrix Equality (LMI) and the observer's gain was calculated. In continuous-time, the concept of auxiliary output was defined, and based on the value of the relative degree, an adaptive high order sliding mode differentiator was proposed to estimate auxiliary output. According to the value of the relative degree, two approach were given, it was shown that the second approach was valid, that is, only a differentiator is needed to estimate states and reconstruct the unknown input. It was proven that the differentiator provides exact estimations under the assumption that states and unknown input and their derivatives are bounded. Even if the bounds are unknown, by the mean of an adaptation law, the differentiator's gains were synthesized. Finally, simulation results were carried out to shown the effectiveness of proposed algorithm

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A. Appendix A

A.1 Stability in the sense of Lyapunov

A.1.1 Fundamental notion of stability

The notion of stability for a dynamical system constitutes a significant problematic of the control theory. It characterizes the evolution of the system's state trajectory initialized around an equilibrium point.

We consider a nonlinear autonomous system (the same analogy is valid for linear systems) of the form

$$\dot{x} = f(x), \tag{A.1}$$

where $x \in \mathbb{R}^n$; $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is continuous.

We designate by $x(t, t_0, x_0)$ the solution of the differential equation (A.1) at the instant $t > t_0$ initialized in $x(t_0) = x_0$.

Equilibrium point physically, a system is in equilibrium if it conserves its state in the absence of external disturbances. Mathematically, this is equivalent to say that the time derivative of its state vector is null. The state x_e is an equilibrium point of the system (A.1) if $f(x_e) = 0$; we suppose in the sequel that $x_e = 0$(it could be tracking error or state estimation).

Definition A.1. (Intuitive notion of stability) If every evolution of a system from a neighborhood sufficiently small of an equilibrium point x_e remains at the neighborhood of this point, hence x_e is said to be stable in the sense of Lyapunov.

Definition A.2. (Stability in the sense of Lyapunov) The equilibrium point $x_e = 0$ is stable in the sense of Lyapunov, if for every $R > 0$, there exists $\mathrm{r} > 0$ such that

$$\|x_0\| < \mathrm{r} \rightarrow \|x(t, t_0, x_0)\| < R; \forall t \geq 0,$$

Otherwise, we say the equilibrium is unstable.

Definition A.3. (Attractivity) We say that the origin $x_e = 0$ is

- An attractive equilibrium point, if there exists $\mathrm{r}_0 > 0$, such that

$$\|x_0\| < \mathrm{r}_0 \rightarrow \|x(t, t_0, x_0)\| \rightarrow 0; \text{ when } t \rightarrow \infty$$

- A globally attractive equilibrium point if:

$$\forall x_0 \in \mathbb{R}^n \rightarrow \|x(t, t_0, x_0)\| \rightarrow 0; \text{ when } t \rightarrow \infty$$

Definition A.4 (Asymptotic stability) The equilibrium point $x_e = 0$ is :

- An asymptotically stable equilibrium point, if it is stable and attractive
- A globally asymptotically stable equilibrium point, if it is stable and globally attractive.

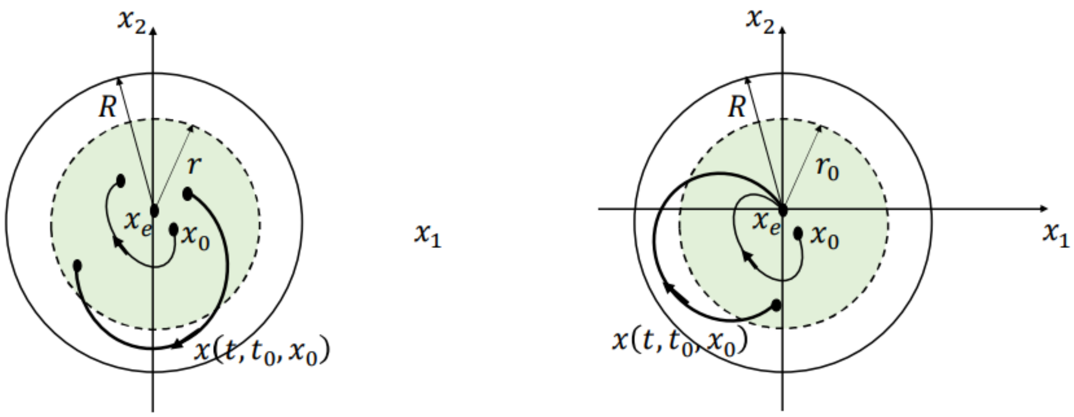

Figure 16. Stability in large sense and asymptotic stability.

Definition A.5. (Exponential stability) The equilibrium point $x_e = 0$ is exponentially stable, if there exist positive constants α, β_1, β_2 such that

$$\|x_0\| < \alpha \rightarrow \|x(t, t_0, x_0)\| < \beta_1 \|x_0\| e^{\beta_2 (t-t_0)}; \quad \forall t \geq t_0,$$

the constant β_2 is called the ratio or convergence speed. If in addition, $r \to \infty$ $\forall x_0 \in \mathbb{R}^n$,$x_e = 0$ is a globally exponentially stable.

Remark: The exponential stability property of the system necessarily induces asymptotic stability.

A.1.2 Direct method of Lyapunov

The direct method of Lyapunov allows to analyze the stability of a system in the neighborhood of its equilibrium point without computing explicitly the solution $x(t, t_0, x_0)$ of the differential equation of the system. This method follows from the concept of energy of a system. For physical systems, energy is a positive definite function of its state. If the system is conservative, the energy remains constant (stable system); it decreases for a dissipative system (asymptotically stable system). If it increases, the system is unstable.

Definition A.6. (Lyapunov function) We call a Lyapunov function for the system (A.1) a real, differentiable and continuous function $V(x)$ that verifies the following properties:

- $V(x) > 0$; $\forall x \neq 0$;
- $V(x) = 0 \rightarrow x = 0$;
- $\dot{V}(x) = \frac{\partial V(x)}{\partial x} f(x) \leq 0; \forall x \neq 0.$

Theorem A.1. If the system admits a Lyapunov function, hence the origin $x_\mathrm{e} = 0$ is a stable equilibrium point. If in addition $\dot{V}(x) = \frac{\partial V(x)}{\partial x} f(x) < 0; \forall x \neq 0$, hence the origin $x_\mathrm{e} = 0$ is an asymptotically stable equilibrium point.

Theorem A.2. The equilibrium point $x_\mathrm{e} = 0$ is exponentially stable if the system admits a Lyapunov function and positive constants η, ρ_1, ρ_2, such that:

- $\rho_1 \|x\|^2 \leq V(x) \leq \rho_2 \|x\|^2$
- $\dot{V}(x) \leq -\eta V(x)$

B. Appendix B

B.1 Observability of linear-time invariant systems

Let us consider the following multi-input-multi-output linear-time invariant system:

$$\begin{cases} \dot{x}(t) = Ax(t) + Bu(t) \\ y(t) = Cx(t) + Du(t) \end{cases} \qquad x(0) = x_0, \tag{B.1}$$

where:

$x(t) \in \mathbb{R}^n$ is the state variable, $u(t) \in \mathbb{R}^m$ is the system vector input and $y(t) \in \mathbb{R}^p$ is the system output. We define $A \in \mathbb{R}^{nxn}$as the dynamic matrix, $B \in \mathbb{R}^{nxm}$ the input matrix, $C \in \mathbb{R}^{pxn}$ output matrix and $D \in \mathbb{R}^{pxm}$ is the transmission matrix.

Definition B.1. (Observability) The pair (C, A) is observable on $[0, T]$ if, given $y(t)$ for $t \in [0, T]$, we can find x_0

Theorem B.1. The following are equivalent

i) (C, A) is observable
ii) The observability matrix $O_{CA} = [C^T \ (CA)^T \ (CA^2)^T \ \ldots \ (CA^{n-1})^T]^T$ has full column rank.
iii) The matrix $\begin{bmatrix} \lambda I_n - A \\ C \end{bmatrix}$ has full column rank for all $\lambda \in \mathbb{C}$.
iv) The eigenvalues of $A + LC$ can be freely assigned by a suitable choice of L.

B.2 Detectability of linear-time invariant systems

Definition B.2. (Detectability) The system (B.1) or the pair (C, A) is detectable if $A + LC$ is stable for some L.

Theorem B.2. The following are equivalent:

i) (C, A) is detectable.
ii) The matrix$\begin{bmatrix} \lambda I_n - A \\ C \end{bmatrix}$ has full column rank for all $Re(\lambda) \geq 0$.
iii) There exists a matrix L such that $A + LC$ is Hurwitz.

B.2.1 Strong detectability and observers

We will introduce the concept of strong detectability of continuous-time system.

Definition B.3. (Strong Detectability) The system (B.1) is **strongly detectable** if

$$y(t) = 0 \quad \text{for } t > 0 \text{ implies } x(t) = 0 \ (t \longrightarrow \infty)$$

For all inputs and initial states.

Remark: As we will see next, **strong detectability** and **strong* detectability** are different concepts.

Definition B.4. (Strong* detectability) The system (B.1) is strong* detectable if

$$y(t) \rightarrow 0 \quad \text{for } t \rightarrow \infty \text{ implies } x(t) \rightarrow 0 \ (t \longrightarrow \infty)$$

Before defining the algebraic conditions for strong and strong* detectability. We will see that strong detectability corresponds to the minimum-phase condition i.e., zeros of the systems are stable. We define the zeros of the system to be the values of $s \in \mathbb{C}$ for which

$$rank \begin{bmatrix} sI_n - A & -B \\ C & D \end{bmatrix} < n + rank \begin{bmatrix} -B \\ D \end{bmatrix}$$

Theorem B.3. The system (B.1) is strongly detectable if and only if all its zeros s satisfy $Re(s) < 0$.

Theorem B.4. The system (B.1) is strong* detectable if and only if it is strong detectable and in addition

$$rank \begin{bmatrix} CB & D \\ D & 0 \end{bmatrix} = n + rank \begin{bmatrix} B \\ D \end{bmatrix} \tag{B.2}$$

In the context of strong detectability, there exists another observability property which is defined as strong observability.

Theorem B.5. The system (B.1) is strongly observable if and only if all it has no zeros.

Definition B.5. (Strong observer) A system Σ with input $y(t)$ and output $\hat{x}(t)$ is called a strong observer for (B.1) if for each input and for each initial state of the original system A.1 as well as of the system Σ, we have

$$x(t) - \hat{x}(t) \rightarrow 0 \qquad (t \longrightarrow \infty)$$

The observer is called strong because it gives an asymptotically correct estimate of the state of the system (B.1) based on its output only. A necessary and sufficient condition for the existence of such an observer is the following

Theorem B.6. The system (B.1) has a strong observer if and only if it is strong* detectable.
Let's consider the following linear discrete-time system

$$\begin{cases} x_{k+1} = Ax_k + Bu_k \\ y_k = Cx_k + Du_k \end{cases} \quad x_0 \; known \tag{B.3}$$

Remark: In discrete-time system, strong detectability and strong* detectability are equivalent.

Theorem B.7. The system (B.3) is strongly detectable if and only if all zeros z satisfy $|z| < 1$.

B.3 Observer gain calculation and estimation error convergence proof

Let us consider the system 3.6,3.7 and observer 3.8:

$$\begin{cases} z_{k+1} = A_d z_k + B_d \bar{u}_k + F_d \bar{\mu}_k - W_d v_k + \Delta_d \\ y_k = \bar{C}_d z_k \end{cases},$$

$$\begin{cases} E\xi_{k+1} = A_\xi \xi_k + B_\xi \bar{u}_k + F_\xi \bar{\mu}_k + \Delta_\xi \\ y_k = C_\xi \xi_k \end{cases},$$

$$\begin{cases} w_{k+1} = Nw_k + Ly_k + M\bar{u}_k + G\bar{\mu}_k + P_z \Delta_\xi \\ \hat{\xi}_k = w_k + Q_z y_k \end{cases},$$

$z_k \in \mathbb{R}^n$ is the state variable, $\bar{u}_k \in \mathbb{R}^m$, $\bar{\mu}_k \in \mathbb{R}^m$ is the system vector inputs and $v_k \in \mathbb{R}^{n_f}$ is the system unknown input,$y(t) \in \mathbb{R}^p$ is the system output. We define $A_d \in \mathbb{R}^{nxn}$as the dynamic matrix, $B_d \in \mathbb{R}^{nxm}$,$F_d \in \mathbb{R}^{nxm}$, $\Delta_d \in \mathbb{R}^{nxm}$ are the input matrices and $W_d \in \mathbb{R}^{nxn_f}$ is the fault distribution matrix and $\bar{C}_d \in \mathbb{R}^{pxn}$ output matrix. We define also $E = [I_\text{n}\ W_d]$, $A_\xi = [A_d\ 0]$, $B_\xi = [B_d^T\ 0]^T$, $F_\xi = [F_d^T\ 0]^T$, $\Delta_\xi = [\Delta_d^T\ 0]^T$and $C_\xi = [\bar{C}_d\ 0]$.

Before proceeding to error convergence analysis, we assume the following hypothesis

- Rank of $E < n + n_f$.
- Rank of $\bar{C}_d = p$.
- Rank of $\begin{bmatrix} E \\ C_\xi \end{bmatrix} = n + n_f$.

As mentioned above, our goal is to design Observer matrices N, L, M, G, P_z, and Q_z. We define the estimation error

$$\tilde{\xi}_k = \xi_k - \hat{\xi}_k = \xi_k - w_k + Q_z C_\xi \xi_k = \left(I_4 - Q_z C_\xi\right)\xi_k - w_k, \tag{B.4}$$

we consider the following condition

$$P_z E + Q_z C_\xi = I_n, \tag{B.5}$$

thus, we obtain

$$\tilde{\xi}_k = P_z E \xi_k - w_k, \tag{B.6}$$

the dynamic of the error is given by

$$\begin{aligned} \tilde{\xi}_{k+1} &= P_z E \xi_{k+1} - w_{k+1} \\ &= P_z A_\xi \xi_k + B_\xi \bar{u}_k + F_\xi \bar{\mu}_k + \Delta_\xi - (N w_k + L y_k + M \bar{u}_k + G \bar{\mu}_k + \ P_z \Delta_\xi) \\ &= P_z A_\xi \xi_k + B_\xi \bar{u}_k + F_\xi \bar{\mu}_k + \Delta_\xi - \left(N w_k + L y_k + M \bar{u}_k + G \bar{\mu}_k + \ P_z \Delta_\xi\right), \end{aligned} \tag{B.7}$$

using the observer output, we write

$$\tilde{\xi}_{k+1} = P_z A_\xi \xi_k + (P_z B_\xi - M)\bar{u}_k + (P_z F_\xi - G)\bar{\mu}_k - N(\hat{\xi}_k - Q_z y_k) - L y_k, \tag{B.8}$$

if we substitute the (3.8) in the equation above, we obtain

$$\begin{aligned} \tilde{\xi}_{k+1} &= P_z A_\xi \xi_k + (P_z B_\xi - M)\bar{u}_k + (P_z F_\xi - G)\bar{\mu}_k + (N Q_z C_\xi - L C_\xi)\xi_k \\ \tilde{\xi}_{k+1} &= -N\left(\xi_k - \tilde{\xi}_k\right) + P_z A_\xi \xi_k + (P_z B_\xi - M)\bar{u}_k + (P_z F_\xi - G)\bar{\mu}_k + (N Q_z C_\xi - L C_\xi)\xi_k \\ \tilde{\xi}_{k+1} &= N\tilde{\xi}_k + (P_z A_\xi - N + N Q_z C_\xi - L C_\xi)\xi_k + (P_z B_\xi - M)\bar{u}_k + (P_z F_\xi - G)\bar{\mu}_k, \end{aligned} \tag{B.9}$$

using (B.5), we obtain

$$\tilde{\xi}_{k+1} = N\tilde{\xi}_k + (P_z A_\xi + N P_z E - L C_\xi)\xi_k + (P_z B_\xi - M)\bar{u}_k + (P_z F_\xi - G)\bar{\mu}_k, \tag{B.10}$$

we can always choose L such that

$$L = -K + N Q_z, \tag{B.11}$$

where K is a matrix of appropriate dimension. Substituting (B.11) into the error equation above we get

$$\tilde{\xi}_{k+1} = N\tilde{\xi}_k + (P_z A_\xi - N + KC_\xi)\xi_k + (P_z B_\xi - M)\bar{u}_k + (P_z F_\xi - G)\bar{\mu}_k.$$

The observer (3.8) will estimate asymptotically ξ_k in other words $\tilde{\xi}_k$ tends to zero asymptotically, if the following conditions are satisfied

i) N is a Hurwitz matrix, i.e., if λ is the eigenvalue of N then we have $\forall\ \lambda \in \mathbb{C}, |\lambda| < 1$.
ii) $N = P_z A_\xi - KC_\xi$.
iii) $P_z B_\xi = M$.
iv) $P_z F_\xi = G$.

Further we can rewrite (B.5) as follows

$$[P_z\ Q_z]\begin{bmatrix} E \\ C_\xi \end{bmatrix} = I_n \tag{B.12}$$

We define the generalized inverse for a given non-square matrix

$$X^+ = X^T(XX^T)^{-1}, \tag{B.13}$$

therefore, we can compute P_z and Q_z using the following formula

$$[P_z\ Q_z] = \begin{bmatrix} E \\ C_\xi \end{bmatrix}^+ = \begin{bmatrix} E \\ C_\xi \end{bmatrix}^T \left(\begin{bmatrix} E \\ C_\xi \end{bmatrix}\begin{bmatrix} E \\ C_\xi \end{bmatrix}^T\right)^{-1}, \tag{B.14}$$

if all the above conditions are satisfied, the error dynamic is given by

$$\tilde{\xi}_{k+1} = N\tilde{\xi}_k,$$

B.4 Lyapunov Analysis

Let us consider the discrete-time candidate Lyapunov function

$$V_k = \tilde{\xi}_k{}^T P\tilde{\xi}_k, \tag{B.15}$$

where V_k is a positive definite function, if its derivative is negative along the trajectories of the system then this indicates stability and P is a symmetric positive definite matrix that we want to find. We have

$$\Delta V_k = V_{k+1} - V_k, \tag{B.16}$$
$$\Delta V_k = \tilde{\xi}_{k+1}{}^T P\tilde{\xi}_{k+1} - \tilde{\xi}_k{}^T P\tilde{\xi}_k, \tag{B.17}$$
$$= \tilde{\xi}_k{}^T (P_z A_\xi - KC_\xi)^T P(P_z A_\xi - KC_\xi)\tilde{\xi}_k - \tilde{\xi}_k{}^T P\tilde{\xi}_k, \tag{B.18}$$
$$= \tilde{\xi}_k{}^T [(P_z A_\xi - KC_\xi)^T P(P_z A_\xi - KC_\xi) - P]\tilde{\xi}_k, \tag{B.19}$$

the objective is to find K and P such that the following quantity is negative

$$(P_z A_\xi - KC_\xi)^T P(P_z A_\xi - KC_\xi) - P < 0, \tag{B.20}$$

if (B.20) is holds, we will obtain

$$\Delta V_k < 0, \tag{B.21}$$

hence, V_k is decreasing over time, which means the origin $\tilde{\xi}_k = 0$ is asymptotically stable.

- Calculation of K

The gain K and the Lyapunov P will be chosen so that the matrix inequality (B.20) is satisfied.

We develop (B.20), we write

$$[(P_zA_\xi)^TP-(KC_\xi)^TP][P_zA_\xi-KC_\xi]-P<0, \tag{B.21}$$

$$(P_zA_\xi)^TP(P_zA_\xi)-(KC_\xi)^TPP_zA_\xi-(P_zA_\xi)^TPKC_\xi+(KC_\xi)^TPKC_\xi-P<0, \tag{B.22}$$

defining $A_z=P_zA_\xi$ and $X=PK$,

$$A_z{}^TPA_z-(XC_\xi)^TP_zA_\xi-(P_zA_\xi)^TXC_\xi+(KC_\xi)^TPP^{-1}PKC_\xi-P<0 \tag{B.23}$$

$$A_z{}^TPA_z-(XC_\xi)^TA_z-A_z{}^TXC_\xi-P+(XC_\xi)^TP^{-1}XC_\xi<0 \tag{B.24}$$

$$A_z{}^TPA_z-(XC_\xi)^TA_z-A_z{}^TXC_\xi-P+(XC_\xi)^TP^{-1}XC_\xi<0, \tag{B.25}$$

we define the Schur's complement as follow:

Let M be a symmetric n x n matrix written as a 2 x 2 block matrix

$$M=\begin{bmatrix} Y & R \\ R^T & H \end{bmatrix},$$

where Y is a p x p matrix and D is a q x q matrix, with $n=p+q$.

If H is invertible then the following properties hold

1) $M>0$ if and only if $H>0$ and $Y-RH^{-1}R^T>0$.
2) If $C>0$, then $M\geq 0$ if and only if $Y-RH^{-1}R^T\geq 0$.

By applying Schur complement to (B.25), we obtain the following LMI

$$S=\begin{bmatrix} A_z{}^TPA_z-A_z{}^TXC_\xi-C_\xi{}^TX^TA_z-P & XC_\xi \\ (XC_\xi)^T & -P \end{bmatrix}<0, \tag{B.26}$$

this LMI is solved numerically using YALMIP and SeDuMI tools in MATLAB. Simulation results are summarized in the following table

Poles	Observer's gain
$[0,0,0.001,0.4995]^T$	$\begin{bmatrix} 0.2502 & -0.005 \\ -0.0018 & 1.0000 \\ 0.2189 & -99.9950 \\ -9.6978 & 2757.1 \end{bmatrix}$

Table 3. Assigned poles and gains of UIO.

C. Appendix C

C.1 Generality about sliding modes

The control by sliding modes is classified into two major categories:
- First or (Classical)Order Sliding Mode Control.
- High Order Sliding Mode Control.

For the sake of illustration of the concept of sliding mode control, we will firstly consider the first order sliding mode control and move afterwards to control by high order sliding modes.

C.1.1 First Order Sliding Mode Control

Let us consider a nonlinear system

$$\dot{x}(t) = f(x, u, t), \tag{C.1}$$

with $x = [\, x_1\, x_2 \dots x_n]^T \in \mathbb{R}^n$ the system's state and $u \in \mathbb{R}^n$ the control input. The synthesis problem of control by sliding modes is done in two steps: First, the sliding surface defined by

$$S = \{x \in \mathbb{R}^n / \, s(x, t) = 0\}, \tag{C.2}$$

fixing the desired dynamic of sliding, is determined. This surface translates the trajectory to impose by the control of the state of (C.1).

Definition C.1. (Sliding phase) We said that there exists an ideal sliding phase of first order on s if there exists a finite time t_s such that the solution of (C.1) satisfies $s(x, t) = 0$ for all $t > t_s$.

Secondly, a discontinuous control law is being calculated. This one must make the surface (C.2) attractive, which is equivalent to constraining the state of the system to reach and remain on the surface S. The sliding mode control law consists in synthesizing a discontinuous control law defined by

$$\begin{cases} u^+(x, t) & \text{if} \quad s(x, t) > 0 \\ u^-(x, t) & \text{if} \quad s(x, t) < 0 \end{cases}, \tag{C.3}$$

where $u^+(x, t)$ and $u^-(x, t)$ are continuous functions. The function $s(x, t)$, that defines the sliding surface, is a continuous function that divides the state space in two regions:
$\{x \in \mathbb{R}^n \,/\, s(x, t) > 0\}$ and $\{x \in \mathbb{R}^n \,/\, s(x, t) < 0\}$.
This can be a linear or nonlinear function and must be properly chosen according to the problem.

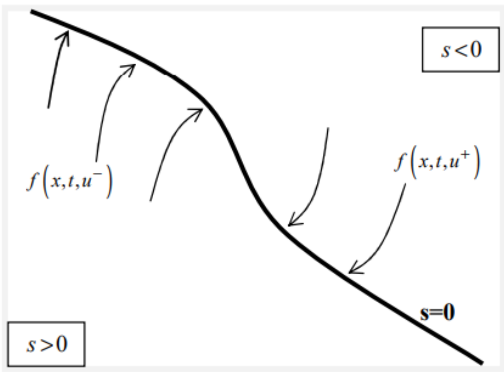

Figure 17. Attractivity of the sliding surface in the phase plan (s, ṡ)

The attractivity of the sliding surface is ensured if and only if the vectors $f(x,u^{+},t)$ and $f(x,u^{-},t)$ are oriented towards each other, hence the assurance of the appearance of the sliding mode. The existence of sliding phase, the surface S must be attractive in every point in the state space. This is ensured if, at neighborhood of the surface S, s and its time derivative $\dot{s}$ are of opposed signs. The attractivity of S, hence the existence of the sliding phase, for (C.2), can be studied by the direct method of Lyapunov for the stability of nonlinear systems. Under some hypotheses, it is sufficient so to find a function $V(s)$ positive definite whose derivative $\dot{V}$ is such that $\dot{V}(s) < 0$,$\forall\, s \neq 0$.

These hypotheses being satisfied in the case of a quadratic function, this function is generally defined by $V(s) = \frac{1}{2}s^2$ and for the appearance of sliding phases, it is sufficient that $\dot{V}(s) < 0$, $\forall s \neq 0$ which guarantees $s\dot{s} < 0$.

In the following, we are interested from now on at the synthesis of control by sliding modes control. For this, we consider the system (C.1) written in the following form:

$$\dot{x}(t) = f(x,t) + g(x,t)u \tag{C.4}$$

C.1.2 Synthesis of first order sliding mode control

For the synthesis of classical sliding modes control, we introduce in the following the method of equivalent command. We suppose that the initial condition x_0 of (C.4) such that $s(x_0\,, t_0\,) = 0$ and that the evolution of the system is maintained on this surface such that $s(x\,, t\,) = 0$ for $t > t_0$.

Theorem C.1. A first order sliding phase exists on the surface S if and only if (C.4) has a relative degree one with respect to the sliding variable s.

The synthesis of the control law by the method of equivalent command, and in the case of the existence of first order sliding mode, the system is admitted being controlled by the equivalent control u_{eq} that ensures the invariance of the surface in time such that $s\big(x, u_{eq}\,, t\,\big) = 0$ and $\dot{s}\big(x, u_{eq}\,, t\,\big) = 0$.

For the synthesis of this command, the derivative of the sliding surface equation is given by

$$\dot{s}(x,t\,) = \frac{\partial s(x,t)}{\partial x}\dot{x} + \frac{\partial s(x,t)}{\partial t} = \frac{\partial s(x,t)}{\partial x}\big(f(x) + g(x)u_{eq}\big) + \frac{\partial s(x,t)}{\partial t} = 0 \tag{C.5}$$

If we suppose the term $\frac{\partial s(x,t)}{\partial x}g(x)$ is non-singular $\forall x \in \mathbb{R}$, the solution u_{eq} of the (C.5) is so given by

$$u_{eq} = \left(\frac{\partial s}{\partial x}g(x)\right)^{-1}\left(\frac{\partial s}{\partial x}f(x) + \frac{\partial s}{\partial t}\right),$$

This command describes the ideal sliding, in absence of disturbance, and guarantees therefore the attractiveness of the surface. To give robustness character to this command, a discontinuous loop is added to the equivalent command in order to assure the convergence to the sliding surface and to counteract the effects of disturbances. Therefore, the first order sliding modes control is composed of the equivalent command u_{eq} and a discontinuous term. We have this case:

$$u = u_{eq} - \eta \left(\frac{\partial s}{\partial x} g(x) \right)^{-1} sign(s), \tag{C.6}$$

with $\eta > 0$ is a constant and the function $sign(.)$ is defined by

$$sign(x) = \begin{cases} 1 & if \quad x > 0 \\ 0 & if \quad x = 0 \\ -1 & if \quad x > 0 \end{cases} \tag{C.7}$$

Control techniques by classical sliding modes suffer, despite their simplicity and robustness, of a major drawback which is the inherent *Chattering* phenomenon. For more details see [23].

C.1.3 Synthesis of high order sliding mode control

The approach of classical sliding mode control techniques is a method known for its robustness with regards to certain types of uncertainties. However, it suffers from the inherent disadvantage earlier spoken of. Alternative solution of classical sliding modes is based on the theory of high order sliding mode techniques. In this approach, in the synthesized command, the discontinuous term is no longer appearing directly in this one but in one of its high order derivatives which is worth to reduce the *Chattering*. Further details of high order sliding mode techniques are found in [23].

C.2 Application of sliding mode on observers

In this section, we will discuss directly high order sliding mode observer and not classical ones. A high order sliding mode observer is based on a recursive form allowing to estimate the n-th derivative of a measured signal.

Let us consider the dynamic system defined by the state representation

$$\begin{cases} \dot{x}_i = x_{i+1} \\ \quad \dots \\ \dot{x}_n = f(x) + \omega \end{cases}, \tag{C.8}$$

where $i = 1, \dots, n-1$, $x = [x_1\ x_2\ \dots\ x_n] \in \mathbb{R}^n$ is the system state, ω is the unknown input, $f(x)$ is a nonlinear function in x and $y = x_1$ represents the measured output whose n-th derivative admits a Lipchitz constant $C > 0$ that can be known.

For implementation of the high order sliding mode observer, we introduce the ensuing hypotheses.

1) The system (C.8) is uniformly bounded and $\forall t > 0\ |x_i(t)| < d_i\ (i = 1, \dots, n)$
2) The unknown input $\omega(t)$ is bounded such that
$$|\omega(t)\ | < \Gamma \text{ and } |\dot{\omega}(t)| < \Gamma'\ \forall t > 0$$
3) $f(x)$ is defined such that there exist constants F and F' verifying

$$f(x(t)) < F \text{ and } \dot{f}(x(t)) < \Gamma'\ \forall t > 0$$

Consider the observer structure defined by

$$\begin{cases} \dot{\hat{x}}_1 = \hat{x}_2 - w_1 - K_1 s_1 \\ \dot{\hat{x}}_2 = \hat{x}_3 - w_2 - K_2 s_2 \\ \quad \dots \\ \dot{\hat{x}}_n = \lambda_{n+1}\hat{x}_{n+1} - w_n - K_n s_n \\ \dot{\hat{x}}_{n+1} = -w_{n+1} \end{cases}, \tag{C.9}$$

with

$$\begin{cases} w_0 = \hat{x}_1 - y \\ w_j = \lambda_j |w_{j-1}|^{\frac{n-j+1}{n-j+2}} sign(w_{j-1}) \end{cases} j = 1, \dots, n+1, \tag{C.10}$$

$\lambda_j > 0$ $(j = 1, \dots, n+1)$ are positive gains representing the tuning parameter of the observer. With adequate choice of λ_j $(j = 1, \dots, n+1)$, the observer (C.9) provides, in real time and in absence of noise, an exact estimation in finite-time of $y(t)$, $\dot{y}(t)$,$\ddot{y}(t)$, …, $y^{(n)}(t)$.The output observer (C.9) is then $\hat{x} = [\hat{x}_1, \hat{x}_2, \dots, \hat{x}_n]$ estimate of the state $[x_1\ x_2\ \dots\ x_n]$ such that $\dot{\hat{x}}_j(t) = y^{(j-1)}(t)$ $(j = 1, \dots, n+1)$.

The parameters λ_j $(j = 1, \dots, n+1)$is difficult to choose and depend on C. In the case where C is known these parameters can be calculated using formula in [18]. When C is unknown and since λ_j $(j = 1, \dots, n+1)$ depend on the accuracy and quality of estimation of measured signal derivative another approach of tuning is by the mean of adaptation.

The observer (C.9) with the following parameter tuning algorithm

$$\begin{cases} w_0 = s_1 \\ w_j = \hat{\lambda}_j |s_j|^{\frac{n-j+1}{n-j+2}} sign(s_j) \end{cases} j = 1, \dots, n+1, \tag{C.11}$$

is an observer for the system (C.8) and provides, in finite time and in the absence of measurement noise, the estimation of derivatives of $y(t)$ to n order.

where the sliding surfaces $s_1 = \hat{x}_1 - y$ and $s_j = \hat{x}_j - \dot{\hat{x}}_{j-1}$ $(j = 1, \dots, n)$, define derivative estimation errors. The adapted parameters in real time according to the adaptive law

$$\frac{d}{dt}\hat{\lambda}_j = \left[|s_j|^{\frac{n-j+1}{n-j+2}} sign(s_j) \right] s_j \quad \text{for} \quad j = 1, \dots, n-1, \tag{C.12}$$

and

$$\frac{d}{dt}\begin{bmatrix} \hat{\lambda}_n \\ \hat{\lambda}_{n+1} \end{bmatrix} = \begin{bmatrix} |s_n|^{\frac{1}{2}} sign(s_n) \\ \int sign(s_n) \end{bmatrix} s_n, \tag{C.13}$$

remark: In the case of multi-output system, the relative degree of each output is computed and the same procedure as in the case of mono-output system is undertook.

C.2.1 Error convergence analysis for adaptive high order sliding mode differentiator

Considering the structure of the observer (C.9) and let λ_j $(j = 1, \dots, n-1)$ be fictive fixed parameters in a way to satisfy the convergence of the observer.
For $j = 1$ we have

$$\dot{\hat{x}}_1 = \hat{x}_2 - \hat{\lambda}_1 |s_1|^{\frac{n}{n+1}} sign(s_1) - K_1 s_1, \tag{C.14}$$

The derivative of $s_1 = \hat{x}_1 - y$ is written

$$\dot{s}_1 = \dot{\hat{x}}_1 - \dot{y}, \tag{C.15}$$

consider λ_1 a fictive fixed parameter such that the expression

$$\dot{y} = \hat{x}_2 - \hat{\lambda}_1 |s_1|^{\frac{n}{n+1}} sign(s_1), \tag{C.16}$$

is verified.

This expression corresponds to the non-adaptive observer (C.9) and this value of λ_1 allows hence the exact estimate of the first derivative of the signal y.

Substituting (C.16) into (C.14) into (C.15), this later is rewritten

$$\begin{aligned}\dot{s}_1 &= \hat{x}_2 - \hat{\lambda}_1 |s_1|^{\frac{n}{n+1}} sign(s_1) - K_1 s_1 - \hat{x}_2 + \hat{\lambda}_1 |s_1|^{\frac{n}{n+1}} sign(s_1) \\ &= \tilde{\lambda}_1 |s_1|^{\frac{n}{n+1}} sign(s_1) - K_1 s_1,\end{aligned} \tag{C.17}$$

where $\tilde{\lambda}_1 = \hat{\lambda}_1 - \lambda_1$.

To prove the convergence, we proceed using direct method of Lyapunov. For this reason, we consider the function

$$V_1 = \frac{1}{2}\left(s_1^2 + \tilde{\lambda}_1^2\right), \tag{C.18}$$

the derivative of $V_1$is given by

$$\begin{aligned}\dot{V}_1 &= s_1 \dot{s}_1 + \dot{\hat{\lambda}}_1 \tilde{\lambda}_1, \\ \dot{V}_1 &= s_1 \left(-\tilde{\lambda}_1 |s_1|^{\frac{n}{n+1}} sign(s_1) - K_1 s_1\right) + \dot{\hat{\lambda}}_1 \tilde{\lambda}_1, \\ \dot{V}_1 &= -K_1 s_1^2 + \tilde{\lambda}_1 \left(-|s_1|^{\frac{n}{n+1}} sign(s_1) s_1 + \dot{\hat{\lambda}}_1\right),\end{aligned} \tag{C.19}$$

given the adaptation law (C.12), and substituting into (C.19), we obtain

$$\dot{V}_1 = -K_1 s_1^2, \tag{C.20}$$

hence, with the choice $K_1 > 0$, we assure estimation error convergence of s_1 such that s_1 tends to 0 therefore, we obtain, in finite time, $\dot{\hat{x}}_1 = \hat{x}_2 = \dot{y}$.
The same procedure is followed for the high order states,$(j = 2, \dots, n-1)$.
For $j = n$, we take the last two rows of (C.9) are written

$$\dot{\hat{x}}_n = -\hat{\lambda}_{n+1} \int sign(s_n)\, dt - \hat{\lambda}_n |s_n|^{\frac{1}{2}} sign(s_n) - K_n s_n, \tag{C.21}$$

The parameters $\lambda_n$and $\lambda_{n+1}$are given such that they verify the equation

$$y^{(n)} = -\lambda_{n+1} \int sign(s_n)\, dt - \lambda_n |s_n|^{\frac{1}{2}} sign(s_n), \tag{C.22}$$

where $s_n = \hat{x}_n - \dot{\hat{x}}_{n-1}$ and we write

$$\dot{s}_n = \dot{\hat{x}}_n - \ddot{\hat{x}}_{n-1}, \tag{C.23}$$

the equality $\dot{\hat{x}}_{n-1} = f^{(n)}$.Therefore, (C.23) becomes

$$\dot{s}_n = \dot{\hat{x}}_n - f^{(n)}, \tag{C.24}$$

substituting (C.21), (C.22) into (C.24), we obtain

$$\dot{s}_n = -\tilde{\lambda}_{n+1} \int sign(s_n)\, dt - \tilde{\lambda}_n |s_n|^{\frac{1}{2}} sign(s_n) - K_n s_n, \tag{C.25}$$

where $\tilde{\lambda}_n = \hat{\lambda}_n - \lambda_n$ and $\tilde{\lambda}_{n+1} = \hat{\lambda}_{n+1} - \lambda_{n+1}$

for the Lyapunov function $V_n = \frac{1}{2}\left(s_n^2 + \tilde{\lambda}_n^2 + \tilde{\lambda}_{n+1}^2\right)$, its derivative $\dot{V}_\mathrm{n}$ is such that

$$\dot{V}_n = s_n \dot{s}_n + \dot{\tilde{\lambda}}_n \tilde{\lambda}_n + \tilde{\lambda}_{n+1} \dot{\tilde{\lambda}}_{n+1}$$

$$\dot{V}_n = -K_n s_n^2 + s_n \left[\dot{\tilde{\lambda}}_n - |s_n|^{\frac{1}{2}} sign(s_n) s_n\right] + \left[\dot{\tilde{\lambda}}_{n+1} - s_n \int sign(s_n)\, dt\right], \tag{C.26}$$

using the (C.13), we obtain

$$\dot{V}_n = -K_n s_n^2, \tag{C.27}$$

hence, the choice $K_n > 0$, allows to guarantee the error convergence such that s_n tends to 0.In other words, we obtain $\dot{\hat{x}}_n = y^{(n)}$.

Conclusion C.1. The adaptive high order sliding mode differentiator is an exact finite time observer for the estimation of successive derivatives of $y(t)$.